
PROFESSIONAL SUMMARY

Full-Stack Software Engineer with **5+ years** of experience **architecting** and **operating high-scale, distributed systems**. Specialized in **event-driven microservices** using **Spring Boot** and **Node.js**, managing **50K+** daily requests and millions of **Kafka-based** transactions, while building **responsive, high-performance interfaces** with **React.js**. Delivered a **50% reduction** in API latency through strategic **caching** and **query-path optimization**. Leverages **AI-driven workflows** to accelerate **system design**, enforce **code quality guardrails**, and integrate **LLM-based features** - significantly increasing **deployment velocity** and **architectural reliability**. Proven track record of **end-to-end ownership** in **AWS/Azure/Kubernetes environments**, with a focus on **fault tolerance**, **observability**, and **long-term system maintainability**.

CERTIFICATIONS

AWS Certified Solutions Architect – Associate | **Amazon Web Services** | April 2026 | bit.ly/pradaap-aws-saa

TECHNICAL SKILLS

Languages: JavaScript, Python, TypeScript, Java, SQL, C++

Frontend: React.js, Next.js, Vue.js, Angular.js, Redux, D3.js, Tailwind CSS, HTML5, CSS3

Backend: Node.js, Spring Boot, Express.js, GraphQL, REST APIs, WebSockets, Microservices Architecture, Distributed Systems

Databases & Caching: MongoDB, MySQL, Oracle, PostgreSQL, Redis, Elasticsearch

Cloud & Infrastructure: AWS, Azure (AKS, Application Insights, Service Bus), Kubernetes, Docker, Terraform

Messaging & Events: Apache Kafka, RabbitMQ, IBM MQ, Event-Driven Architecture

DevOps & Tools: Git, GitLab CI/CD, GitHub Actions, Jenkins, Kibana, Cloudwatch, Prometheus, Grafana

Security & Testing: OAuth 2.0, JWT, Spring Security, RBAC, Jest, JUnit

AI & Machine Learning: Agentic AI, LLM Integration, TensorFlow, ML Model Integration, Predictive Analytics, ETL Pipelines, Data Processing

PROFESSIONAL EXPERIENCE

Full Stack Developer

Hub Group Inc | Jan 2024 - Present

- Architected event-driven Order Management system in collaboration with senior engineers, using **Vue.js**, **Node.js**, **MongoDB**, and **Kafka** on **Azure infrastructure**, implementing pub-sub messaging patterns for order lifecycle events, processing over **2K+ daily shipment orders** across 100+ distribution centers, reducing order processing cycle time by approximately 60% and achieving **99.5% order completion rate**
- Built Rail Invoicing platform using **React.js frontend** with 6 **Node.js/Express.js** microservices, **GraphQL API layer**, and **MongoDB** backend on **Azure** infrastructure, implementing **OAuth 2.0** authentication, **RBAC** security, and **automated invoice generation workflows** processing tariff calculations, drayage costs, and ERP posting for **12K+ monthly rail shipments**, reducing manual invoicing cycle time by approximately 70% and achieving 99.8% invoice accuracy rate.
- Contributed to **10-month migration** from monolithic COBOL legacy systems to containerized **Node.js** microservices on **Azure Kubernetes Service (AKS)** across 12 services, implementing **Docker** containerization, **Kubernetes** deployments with **rolling updates** and **health checks**, and **GitLab CI/CD** pipelines with automated testing, enabling **zero-downtime releases**, independent service scaling, and ~75% faster deployment times from ~4 hours to ~1 hour.
- Engineered **SAAS integration** platform connecting TMS systems (Mcleod, Power44, TriumphPay, DAT) using **RESTful APIs**, custom **ETL pipelines** for payload transformation and data normalization, and **Kafka** message streaming, implementing **webhook** receivers, data validation, **dead-letter queue** handling with retry mechanisms, and automated alerting, automating **25K+ daily order synchronization transactions** across freight brokerage operations and eliminating over 150 manual data transfer hours weekly.
- Designed real-time order failure detection system using Azure Service Bus topics with automated retry workflows implementing **exponential backoff**, **dead-letter queue handling**, and **exception routing logic**, improving order completion rate to over 99% and reducing manual intervention by approximately 80% for production exception handling across 50K+ daily order processing operations.
- Optimized **distributed API architecture** implementing Redis clustering with **multi-tier caching strategies (cache-aside pattern)** and **MongoDB** compound indexing, reducing average **API response time** by **~55% (from ~450ms to ~200ms)** and **P95 latency** by **~60% (~800ms to ~320ms)**, supporting 5x traffic growth over 18 months while maintaining 99.9% API availability for enterprise-scale order processing.
- Implemented **monitoring** and **alerting** infrastructure using **Kibana** dashboards, **Azure** Application Insights, and custom **CloudWatch** metrics, establishing automated alerts for **P95 latency**, error rates, and **throughput anomalies**, reducing **mean time to detection (MTTD)** from ~30 minutes to ~5 minutes and enabling proactive incident response for production systems handling 50K+ daily transactions.

Software Engineer

Cerner Corporation [Oracle Health] | Apr 2021 - Aug 2022

- Built **REST APIs** using **Spring Boot**, JAX-RS, and MyBatis connecting hospital systems via IBM MQ messaging, implementing HL7 FHIR R4 resource endpoints for Patient and Observation data, reducing cross-system data errors by ~75% across 4 hospital networks.
- Optimized **Oracle** and **MongoDB** performance for clinical reporting through **indexing**, **query refactoring**, and **connection pool tuning**, reducing average report generation time from ~6s to under 2s (~67% improvement) for real-time dashboards used by 200+ clinical staff.
- Identified and resolved **Kafka** consumer group lag causing clinical event processing delays; redesigned topic partitioning to distribute load evenly across 8 partitions, improving throughput by **~65% during peak hours validated via monitoring dashboards**.
- Contributed to migrating 2 monolithic EHR modules to microservices on **Oracle Cloud Infrastructure (OCI)** and **AWS ECS**, containerizing with Docker and implementing **GitLab CI/CD** pipelines, reducing release cycle by ~40%.
- Integrated **Cerner Machine Learning Ecosystem** predictive models into patient monitoring workflows, consuming AWS-hosted ML endpoints via REST APIs to display risk scores for 1,000+ daily patient assessments in React dashboards, supporting proactive clinical interventions.

Programmer Analyst

Cognizant | Sep 2020 - Apr 2021

- Developed **Spring Boot** microservices with **MySQL** for **ACID-compliant** transactional data and **MongoDB** for document storage, implementing **OAuth2/JWT** authentication and **RBAC** on a financial platform serving 10K+ daily users across 40+ enterprise clients with 99.8% availability.
- Built **event-driven** data pipeline using **Kafka** and **RabbitMQ** with **Spring Boot** consumers on **AWS**, developing message handlers for high volume (~3M+) **financial transaction processing** with **idempotency checks & dead-letter queue** handling to ensure reliable message delivery.
- Developed **RESTful API** endpoints with **Redis caching integration** and **rate limiting**, supporting **50K+ daily API requests** and **contributing to ~35% average response time** improvement through caching optimizations and custom interceptors.

EDUCATION

Master of Computer Science - Illinois Institute of Technology, USA